Intermolecular Forces Explorer

Basic Info

London Dispersion: Weak forces due to temporary dipoles.

Dipole-Dipole: Attraction between polar molecules.

Hydrogen Bonding: Strong interaction with H and N/O/F.

Ion-Dipole: Attraction between ion and polar molecule.

Select Molecule

--Select--

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--Select--
H_2O
NH_3
CO_2
CH_4
HCl

Sel cule

--Select--

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Select Molecule

HCl

HCl

Force: Dipole-Dipole

Strength: Moderate

Explanation: Polar bond between H and Cl.

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NH_3

NH3

Force: Hydrogen Bonding

Strength: Moderate

Explanation: Nitrogen allows hydrogen bonding but weaker than water.

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H_2O

H2O

Force: Hydrogen Bonding

Strength: Strong

Explanation: Water has H bonded to O, allowing hydrogen bonding.

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CO_2

CO2

Force: London Dispersion

Strength: Weak

Explanation: Non-polar molecule, only dispersion forces.

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Select Molecule

CH_4

CH4

Force: London Dispersion

Strength: Very Weak

Explanation: Symmetrical non-polar molecule.